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LAB 02

Database: Point of Sale System

1. Introduction

In this lab, we designed a Point of Sale (POS) database system. The database simulates a small retail environment where products are categorized, customers place orders, and discounts are applied. The goal is to practice creating tables, defining relationships, inserting sample data, and running queries to extract useful information.

2. Database Structure

The database consists of the following tables:

- **Roles** – Defines user roles (Admin, Customer, Salesman).
- **Categories** – Groups products into categories such as Beverages, Snacks, Dairy, etc.
- **Products** – Stores product details including name, price, and category.
- **Orders** – Records customer orders with date and customer name.
- **Order Items** – Links products to orders with quantities.
- **Discounts** – Stores discount percentages for products.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ products	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ order_items	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ discounts	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ categories	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
5 tables	Sum	50	InnoDB	utf8mb4_general_ci	144.0 KiB	0 B

3. Data Insertion

Sample data was inserted into each table:

- **Categories:** 10 categories (Beverages, Snacks, Dairy, Bakery, Fruits, Vegetables, Frozen, Meat, Seafood, Grains).
- **Products:** 10 products (Coke, Pepsi, Chips, Milk, Bread, Apple, Carrot, Ice Cream, Chicken, Fish).

```
SELECT * FROM `products`
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: | Filter rows:

Extra options

				product_id	name	price	category_id
☐	Edit	Copy	Delete	1	Coke	100.00	1
☐	Edit	Copy	Delete	2	Pepsi	90.00	1
☐	Edit	Copy	Delete	3	Chips	50.00	2
☐	Edit	Copy	Delete	4	Milk	120.00	3
☐	Edit	Copy	Delete	5	Bread	80.00	4
☐	Edit	Copy	Delete	6	Apple	150.00	5
☐	Edit	Copy	Delete	7	Carrot	70.00	6
☐	Edit	Copy	Delete	8	Ice Cream	200.00	7
☐	Edit	Copy	Delete	9	Chicken	500.00	8
☐	Edit	Copy	Delete	10	Fish	600.00	9

☐ Check all | With selected: Edit Copy Delete Export

Orders and Order Items

- **Orders:** 10 orders placed by customers (Ali, Ahmed, Sara, Zain, Ayesha, Usman, Hina, Bilal, Fatima, Omar).
- **Order Items:** Each order linked with products and quantities.

←

Server: 127.0.0.1 » Database: point_of_sale » Table: orders

Browse

Structure

SQL

Search

Insert

Export

Import

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key:

Extra options

←

T

→

order_id

order_date

customer_name

☐

Edit

Copy

Delete

1

2026-01-01

Ali

☐

Edit

Copy

Delete

2

2026-01-02

Ahmed

☐

Edit

Copy

Delete

3

2026-01-03

Sara

☐

Edit

Copy

Delete

4

2026-01-04

Zain

☐

Edit

Copy

Delete

5

2026-01-05

Ayesha

☐

Edit

Copy

Delete

6

2026-01-06

Usman

☐

Edit

Copy

Delete

7

2026-01-07

Hina

☐

Edit

Copy

Delete

8

2026-01-08

Bilal

☐

Edit

Copy

Delete

9

2026-01-09

Fatima

☐

Edit

Copy

Delete

10

2026-01-10

Omar

☐ Check all

With selected:

Edit

Copy

Delete

Export

- **Discounts:** Discounts are applied to products. Each products can have a discount percentage.

Server: 127.0.0.1 » Database: point_of_sale » Table: discounts

Browse Structure SQL Search Insert Export In

✓ Showing rows 0 - 9 (10 total, Query took 0.0004 seconds.)

`SELECT * FROM `discounts``

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 ▾ | Filter rows: Search this table

Extra options

				discount_id	product_id	discount_percent
☐	Edit	Copy	Delete	1	1	10
☐	Edit	Copy	Delete	2	2	5
☐	Edit	Copy	Delete	3	3	15
☐	Edit	Copy	Delete	4	4	8
☐	Edit	Copy	Delete	5	5	12
☐	Edit	Copy	Delete	6	6	7
☐	Edit	Copy	Delete	7	7	6
☐	Edit	Copy	Delete	8	8	20
☐	Edit	Copy	Delete	9	9	10
☐	Edit	Copy	Delete	10	10	5

↑ ☐ Check all | With selected: Edit Copy Delete Export

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	item_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	order_id	int(11)			Yes	NULL			Change Drop More
☐ 3	product_id	int(11)			Yes	NULL			Change Drop More
☐ 4	quantity	int(11)			Yes	NULL			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext
 Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize
 Add 1 column(s) after quantity Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	item_id	10	A	No	
Edit Rename Drop	order_id	BTREE	No	No	order_id	10	A	Yes	
Edit Rename Drop	product_id	BTREE	No	No	product_id	10	A	Yes	

Create an index on 1 columns Go

Activate Windows
Go to Settings to activate Windows

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	product_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	name	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 3	price	decimal(10,2)			Yes	NULL			Change Drop More
☐ 4	category_id	int(11)			Yes	NULL			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext
 Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize
 Add 1 column(s) after category_id Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	product_id	10	A	No	
Edit Rename Drop	category_id	BTREE	No	No	category_id	10	A	Yes	

4. Queries and Results

- **Total Orders**

This query shows the total number of orders placed.

Your SQL query has been executed successfully.

```
-- Total orders SELECT COUNT(*) AS Total_Orders FROM Orders;
```

- **Total Sales per Product**

This query calculates how many units of each product were sold.

✓ Showing rows 0 - 9 (10 total, Query took 0.0007 seconds.)

```
-- Total sales per product SELECT p.name, SUM(oi.quantity) AS total_sold FROM Order_Items oi JOIN Products p ON oi.product_id = p.product_id GROUP BY p.name;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

name	total_sold
Apple	3
Bread	1
Carrot	4
Chicken	2
Chips	5
Coke	2
Fish	1
Ice Cream	1
Milk	2
Pepsi	1

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Activate Windows

- **Orders with Customer Names**

Displays all orders along with customer names and dates.

Showing rows 0 - 9 (10 total, Query took 0.0004 seconds.)

```
-- Orders with customer names SELECT order_id, customer_name, order_date FROM Orders;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	order_id	customer_name	order_date
☐ Edit Copy Delete	1	Ali	2026-01-01
☐ Edit Copy Delete	2	Ahmed	2026-01-02
☐ Edit Copy Delete	3	Sara	2026-01-03
☐ Edit Copy Delete	4	Zain	2026-01-04
☐ Edit Copy Delete	5	Ayesha	2026-01-05
☐ Edit Copy Delete	6	Usman	2026-01-06
☐ Edit Copy Delete	7	Hina	2026-01-07
☐ Edit Copy Delete	8	Bilal	2026-01-08
☐ Edit Copy Delete	9	Fatima	2026-01-09
☐ Edit Copy Delete	10	Omar	2026-01-10

☐ Check all | With selected: Edit Copy Delete Export

Activate Windows
Go to Settings to activate Windows.

- **Products with Discounts**

Shows products with their respective discount percentages.

Showing rows 0 - 9 (10 total, Query took 0.0006 seconds.)

```
-- Products with discounts SELECT p.name, d.discount_percent FROM Products p JOIN Discounts d ON p.product_id = d.product_id;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

name	discount_percent
Coke	10
Pepsi	5
Chips	15
Milk	8
Bread	12
Apple	7
Carrot	6
Ice Cream	20
Chicken	10
Fish	5

These queries demonstrate how the POS database can provide insights into sales and customer activity.

5. ER Diagram Explanation

The Entity-Relationship (ER) Diagram visually represents the structure of the database:

- **Roles:** Independent table, not directly linked to others in this lab.
- **Categories → Products:** One category can have many products (1-to-many relationship).
- **Products → Order_Items:** A product can appear in many order items (1-to-many).
- **Orders → Order_Items:** One order can contain multiple items (1-to-many).
- **Products → Discounts:** Each product can have one discount (1-to-1 relationship in this design).

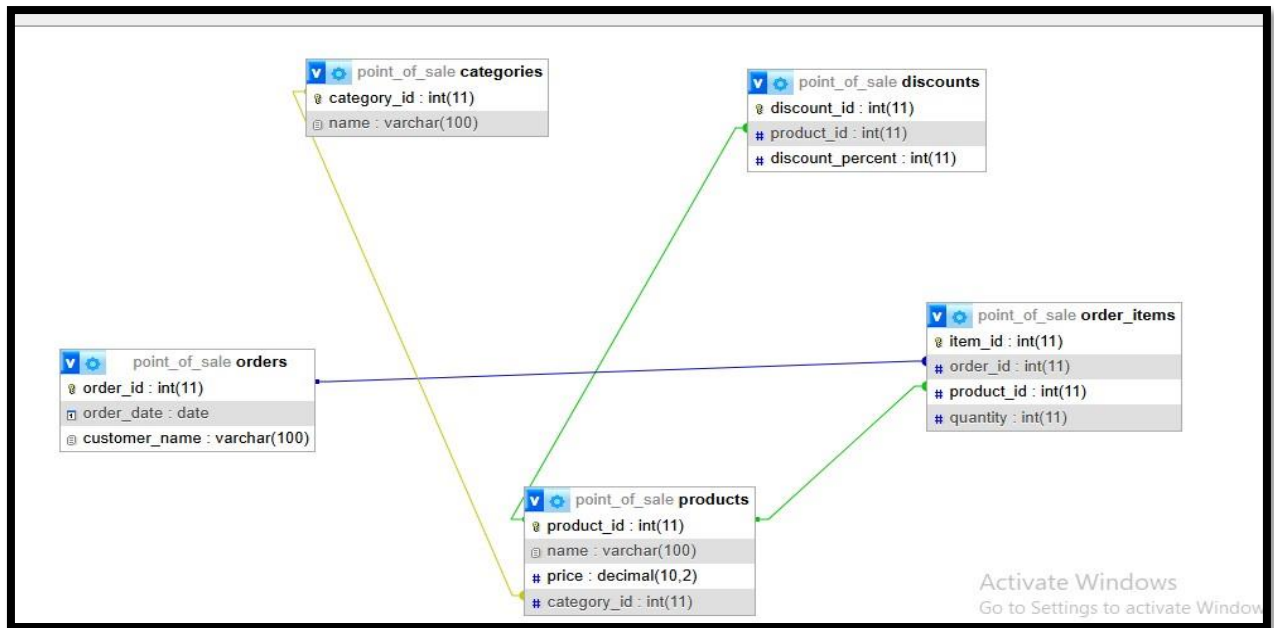
ER Diagram (Conceptual Description)

Entities: Roles, Categories, Products, Orders, Order_Items, Discounts.

Relationships:

- **Categories → Products (1:N)**
- **Orders → Order_Items (1:N)**
- **Products → Order_Items (1:N)**
- **Products → Discounts (1:1)**

This diagram shows how data flows between customers, orders, products, and discounts in the POS system.



Exporting tables from "point_of_sale" database

Export templates:

New template:

Existing templates:

Template name

Template name

Create

Template:

-- Select a template --

Update

Delete

Export method:

☒ Quick - display only the minimal options

☐ Custom - display all possible options

Format:

SQL

Export

Activate Windows
Go to Settings to activate Windows.

6. Conclusion

This lab successfully demonstrates how to build a Point of Sale database with proper relationships between tables. The queries provide meaningful business insights such as total orders, product sales, customer details, and discounts. The ER diagram further clarifies the logical structure of the system, making it easier to understand and extend in future projects.

*****THE END*****